Entropy and mutual information

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This page builds on the definitions of entropy, cross-entropy and Kullback–Leibler divergence in *Probability for Data Science*.

Surprisal

Definition 0.1 (Surprisal)**.** The **surprisal** of an outcome x of a discrete random variable X with $\mathbb{P}(X = x) > 0$ is

$$s(x) \stackrel{\text{def}}{=} -\log \mathbb{P}(X = x).$$

With base-2 logarithms, surprisal is also called the **Shannon information content** of the outcome x (MacKay 2003, sec. 2.4).

Definition 0.2 (Bit (shannon))**.** A **bit**, also called a **shannon** (symbol Sh) (International Electrotechnical Commission 2008), is the unit of information when the logarithm in surprisal, and in every quantity built from it, has base 2.
One bit is the surprisal of an outcome with probability 1/2: for an outcome x of a discrete random variable X,

$$\begin{aligned} s(x) = 1 \text{ bit} &\iff -\log_2 \mathbb{P}(X = x) = 1 \\ &\iff \mathbb{P}(X = x) = 2^{-1} = \tfrac{1}{2}. \end{aligned}$$

In words, a bit is the surprisal of the outcome of one binary trial whose two outcomes are equally likely, such as one toss of a fair coin.

Example 0.1 (Counting bits with fair coins)**.** One toss of a fair coin has 2 equally likely outcomes, and one binary digit is enough to record which one happened. Two tosses have 4 outcomes, and two binary digits record them. In general, k tosses have $n = 2^k$ equally

likely outcomes, and $k = \log_2 n$ binary digits record them.
Each of those n outcomes has probability $p = 1/n$, so

$$\begin{aligned} k &= \log_2 n \\ &= \log_2 \frac{1}{p} \\ &= -\log_2 p, \end{aligned}$$

which is the surprisal of the outcome, in bits.
Ten tosses that all land heads have probability 2^{-10}, so

$$\begin{aligned} s(\text{ten heads}) &= -\log_2 2^{-10} \\ &= 10 \text{ bits}. \end{aligned}$$

Definition 0.3 (Nat)**.** A **nat** is the unit of information when the logarithm in surprisal, and in every quantity built from it, is the natural logarithm.
One nat is the surprisal of an outcome with probability $1/\mathrm{e}$: for an outcome x of a discrete random variable X,

$$\begin{aligned} s(x) = 1 \text{ nat} &\iff -\ln \mathbb{P}(X = x) = 1 \\ &\iff \mathbb{P}(X = x) = \mathrm{e}^{-1} = 1/\mathrm{e} \approx 0.368. \end{aligned}$$

Example 0.2 (One fair coin toss in nats)**.** By Example 0.1 with $k = 1$, one toss of a fair coin landing heads has a surprisal of 1 bit. In nats, the same outcome has a surprisal of

$$\begin{aligned} s(\text{heads}) &= -\ln \tfrac{1}{2} \\ &= \ln 2 \\ &\approx 0.693 \text{ nats}. \end{aligned}$$

Remark 0.1 (Converting between bits and nats). For any probability $p \in (0, 1]$, the change-of-base rule for logarithms gives

$$\begin{aligned} -\log_2 p &= \frac{-\ln p}{\ln 2} \\ &\approx 1.443 \times (-\ln p). \end{aligned}$$

So an outcome with a surprisal of 1 nat has a surprisal of $1/\ln 2 \approx 1.443$ bits, and the $\ln 2 \approx 0.693$ nats of Example 0.2 are exactly $\ln 2/\ln 2 = 1$ bit.
A bit as a unit of information is not the same thing as a bit as a binary digit. The name shannon refers only to the unit of information, so it avoids this ambiguity. A binary digit

whose two values are equally likely has a surprisal of exactly 1 bit, whichever value it takes. A binary digit that is 0 nine times in ten does not: the value 0 has a surprisal of $-\log_2 0.9 \approx 0.152$ bits, and the value 1 has a surprisal of $-\log_2 0.1 \approx 3.322$ bits.

Remark 0.2 (Unlikely outcomes carry more information). Surprisal is large when an outcome was unlikely, and it falls to 0 as the outcome becomes certain. Correctly predicting heads on one toss of a fair coin is worth 1 bit; correctly predicting ten heads in a row is worth 10 bits.

Remark 0.3 (Entropy is expected surprisal). The entropy of a discrete random variable X is the expectation of its surprisal:

$$\begin{aligned} H(X) &= \mathbb{E}[-\log \mathbb{P}(X)] \\ &= \mathbb{E}[s(X)]. \end{aligned}$$

So $H(X)$ is the average number of bits (with base-2 logarithms) needed to record an outcome of X when each outcome x costs $s(x)$ bits. The source coding theorem, on a later page of these notes, makes that average precise: no uniquely decodable code does better.

Definition 0.4 (Rarity)**.** The **rarity** of an outcome x of a discrete random variable X is the reciprocal of its probability mass:

$$\mathbb{r}(x) \stackrel{\text{def}}{=} \frac{1}{\mathbb{P}(X = x)}.$$

For a continuous random variable X with density $\mathbb{p}$, the rarity of a value x is the reciprocal of the density:

$$\mathbb{r}(x) \stackrel{\text{def}}{=} \frac{1}{\mathbb{p}(x)}.$$

When the mass or the density is 0, the rarity is ∞.

Example 0.3 (Rarity of the outcomes of a die)**.**
Let X be the roll of a fair six-sided die. Each outcome has probability $1/6$.

```r
rarity <- function(p) 1 / p
p_fair <- 1 / 6
p_loaded_six <- 1 / 2
p_loaded_other <- (1 - p_loaded_six) / 5
rarity_fair <- rarity(p_fair)
rarity_loaded_six <- rarity(p_loaded_six)
rarity_loaded_other <- rarity(p_loaded_other)
rarities <- c(
  fair = rarity_fair,
  loaded_six = rarity_loaded_six,
  loaded_other = rarity_loaded_other
)
rarities
#>         fair   loaded_six loaded_other
#>            6            2           10
```

So each face of the fair die has rarity $\mathrm{r}(x) = 6$: one roll in six.
Suppose the die is loaded so that a six has probability 1/2. Then $\mathrm{r}(6) = 2$. If the other five faces share the rest equally, each has $\mathrm{r}(x) = 10$. The more probable an outcome, the lower its rarity.

Remark 0.4 (Source of the name). Leinster writes that, for a community whose ith species has relative abundance p_i, "$1/p_i$ is a measure of its rarity or specialness" (Leinster 2021, sec. 2.4, p. 52). The name "rarity" for the reciprocal of a probability is borrowed from there.

Theorem 0.1 (Surprisal is the logarithm of the rarity)**.** *Let X be a discrete random variable, and let x be an outcome with $\mathbb{P}(X = x) > 0$. Then the surprisal of x is the logarithm of its rarity:*

$$\mathrm{s}(x) = \log\{\mathrm{r}(x)\}.$$

Proof

Proof.

$$\begin{aligned}
\mathrm{s}(x) &= -\log\{\mathbb{P}(X = x)\} && \text{(definition of surprisal)} \\
&= \log\left\{\frac{1}{\mathbb{P}(X = x)}\right\} && \text{(logarithm of a reciprocal)} \\
&= \log\{\mathrm{r}(x)\}. && \text{(definition of rarity)}
\end{aligned}$$

□

Theorem 0.2 (Surprisal of a continuous value is the logarithm of the rarity)**.** *Let X be a continuous random variable with density $\mathbb{p}$, and let x be a value with $\mathbb{p}(x) > 0$. Then the surprisal of x is the logarithm of its rarity:*

$$\mathbb{s}(x) = \log\{\mathbb{r}(x)\}.$$

Proof

Proof.

$$\begin{aligned} \mathbb{s}(x) &= -\log\{\mathbb{p}(x)\} && \text{(definition of surprisal)} \\ &= \log\left\{\frac{1}{\mathbb{p}(x)}\right\} && \text{(logarithm of a reciprocal)} \\ &= \log\{\mathbb{r}(x)\}. && \text{(definition of rarity)} \end{aligned}$$

□

Corollary 0.1 (Surprisal of a discrete outcome is nonnegative)**.** *Let X be a discrete random variable, and let x be an outcome with $\mathbb{P}(X = x) > 0$. Then the rarity of x is at least 1, with equality exactly when $\mathbb{P}(X = x) = 1$, and the surprisal satisfies*

$$\mathbb{s}(x) \geq 0.$$

Proof

Proof. Since $0 < \mathbb{P}(X = x) \leq 1$, the rarity $\mathbb{r}(x) = 1/\mathbb{P}(X = x)$ is at least 1, with equality exactly when $\mathbb{P}(X = x) = 1$. The logarithm is increasing, and $\log\{1\} = 0$, so by Theorem 0.1, $\mathbb{s}(x) = \log\{\mathbb{r}(x)\} \geq 0$. □

Example 0.4 (Surprisal and rarity of an outcome with probability one quarter)**.** Take an outcome with probability 1/4:

```
p_four <- 1 / 4
rarity_four <- 1 / p_four
surprisal_four <- log2(rarity_four)
surprisal_four
#> [1] 2
```

It has rarity 4 by definition. By Theorem 0.1, its surprisal is the base-2 logarithm of that

rarity, 2 bits.

Remark 0.5 (Reading the rarity of an outcome). A density can exceed 1, so for a continuous random variable the rarity can be below 1, and then the surprisal is negative (Theorem 0.2). For a mass, a rarity of r reads "one in r", and the surprisal is the logarithm of r (Theorem 0.1).

Remark 0.6 (Values of the probability, the rarity and the surprisal). For a discrete random variable, the mass $\mathbb{P}(X = x)$ of an outcome with a mass above 0, its rarity $\mathbb{r}(x)$ and its surprisal $\mathbb{s}(x)$ take the values in the middle column. For a continuous random variable, the density $\mathbb{p}(x)$ takes the place of the mass. At a value x with a density above 0, the rarity $\mathbb{r}(x)$ is the reciprocal of $\mathbb{p}(x)$, and the surprisal $\mathbb{s}(x)$ is the surprisal of a continuous value. Their values are in the right column.

	Mass $\mathbb{P}(X = x)$	Density $\mathbb{p}(x)$
Mass or density	$(0, 1]$	$(0, \infty)$
Rarity $\mathbb{r}(x)$	$[1, \infty)$	$(0, \infty)$
Surprisal $\mathbb{s}(x)$	$[0, \infty)$	$(-\infty, \infty)$

- **Mass.** A mass of 1 gives the smallest rarity, 1, and the smallest surprisal, 0. No mass exceeds 1, the probability of a certain outcome. So no mass gives a rarity below 1 or a surprisal below 0.
- **Density.** A density can be arbitrarily large, as in the uniform example, so the rarity can fall below 1 and the surprisal can fall below 0.
- **Zero.** A mass or density of 0 gives a rarity and a surprisal of ∞.
- **Units.** A mass has no units. A density has the units of $1/x$, so its rarity has the units of x. Rescaling x by a nonzero c shifts its surprisal by $\log\{|c|\}$.

Remark 0.7 (Differential entropy differs from entropy). A density value is not a probability, so the surprisal of a continuous value and the entropy of a continuous random variable (the differential entropy) behave differently from their discrete counterparts:

- **Sign.** A density can exceed 1, so the surprisal $\mathbb{s}(x)$ and the differential entropy can be negative, as in the uniform example (Cover and Thomas 2006, sec. 8.1). The entropy of a discrete random variable is never negative.
- **Units.** Rescaling changes the differential entropy: $\mathbb{H}(aX) = \mathbb{H}(X) + \log|a|$ for a constant $a \neq 0$ (Cover and Thomas 2006, Theorem 8.6.4). Measuring the same quantity in metres rather than centimetres therefore changes it. Adding a constant

does not: $\mathbb{H}(X + c) = \mathbb{H}(X)$ (Cover and Thomas 2006, Theorem 8.6.3).

- **Not a limit of entropy.** Round a continuous X to bins of width Δ, which gives a discrete random variable X^Δ. Then $\mathbb{H}(X^\Delta) + \log \Delta \to \mathbb{H}(X)$ as $\Delta \to 0$ (Cover and Thomas 2006, Theorem 8.3.1). The entropy $\mathbb{H}(X^\Delta)$ itself grows without bound as $\Delta \to 0$, so the differential entropy is not its limit.

Definition 0.5 (Collision probability)**.** The **collision probability** of a discrete random variable X is the chance that two independent copies X and X' of X take the same value:

$$\mathbb{P}(X = X') = \sum_x \mathbb{P}(X = x)^2.$$

Definition 0.6 (Rényi entropy of order 2 (collision entropy))**.** The **Rényi entropy of order 2**, also called the **collision entropy** (Wikipedia contributors 2026), of a discrete random variable X is the negative logarithm of the expectation of its probability, where $\mathbb{P}(X)$ is the probability mass function evaluated at the random variable X itself:

$$H_2(X) = -\log \mathbb{E}[\mathbb{P}(X)].$$

Theorem 0.3 (The expected probability is the collision probability)**.** *Let X be a discrete random variable, and let X' be an independent copy of X. Then*

$$\mathbb{E}[\mathbb{P}(X)] = \sum_x \mathbb{P}(X = x)^2 = \mathbb{P}(X = X').$$

Proof

Proof. The expectation of $\mathbb{P}(X)$ adds, over the values x, the value $\mathbb{P}(X = x)$ times its probability $\mathbb{P}(X = x)$:

$$\begin{aligned}\mathbb{E}[\mathbb{P}(X)] &= \sum_x \mathbb{P}(X = x)\,\mathbb{P}(X = x) \\ &= \sum_x \mathbb{P}(X = x)^2.\end{aligned}$$

By the definition of the collision probability, the last sum is $\mathbb{P}(X = X')$. □

Theorem 0.4 (The collision entropy is at most the Shannon entropy)**.** *Let X be a discrete random variable with values $x_1, \ldots, x_n$. Then the collision entropy is at most the entropy:*

$$H_2(X) \leq \mathbb{H}(X).$$

The two coincide exactly when each value that X takes with positive probability has the same probability.

Proof

Proof. Write $\mathbb{P}(X)$ for the probability mass function evaluated at X. The function $-\log$ is strictly convex, so Jensen's inequality gives

$$\mathbb{E}[-\log \mathbb{P}(X)] \geq -\log \mathbb{E}[\mathbb{P}(X)],$$

with equality exactly when $\mathbb{P}(X)$ is constant. The left side is $\mathbb{H}(X)$. The right side is $H_2(X)$ by its definition. The function $\mathbb{P}(X)$ is constant exactly when every value that X takes with positive probability has that same probability. □

Example 0.5 (Collision entropy and entropy of a three-valued random variable)**.** Let X take three values with probabilities $1/2$, $1/4$ and $1/4$. The code below computes the expected probability, the collision entropy and the entropy, with base-2 logarithms.

```
p <- c(1 / 2, 1 / 4, 1 / 4)
expected_prob <- sum(p^2)
collision_entropy <- -log2(expected_prob)
shannon_entropy <- -sum(p * log2(p))
c(
  expected_prob = expected_prob,
  collision_entropy = collision_entropy,
  shannon_entropy = shannon_entropy
)
#>     expected_prob collision_entropy   shannon_entropy
#>          0.375000          1.415037          1.500000
```

By Theorem 0.3, the expected probability $\mathbb{E}[\mathbb{P}(X)] = 0.375$ is the collision probability. The collision entropy is $H_2(X) = 1.415$ bits. The entropy is $\mathbb{H}(X) = 1.5$ bits. As Theorem 0.4 states, $H_2(X) \leq \mathbb{H}(X)$, and the two differ because the three values are not equally likely.

Remark 0.8 (Two ways to summarize a random variable by its probabilities). Entropy is the expectation of the surprisal $-\log \mathbb{P}(X)$. Taking the expectation of the probability $\mathbb{P}(X)$ itself summarizes X differently. For a discrete random variable, Theorem 0.3 identifies that expectation with the collision probability, so the collision entropy is $H_2(X) = -\log \mathbb{P}(X = X')$.

For a continuous random variable with density $\mathbb{p}$, the analogue is $\mathbb{E}[\mathbb{p}(X)] = \int \mathbb{p}(x)^2\,dx$. The expected-density exercises derive this quantity, then compute it in the uniform case and the normal case.
Theorem 0.4 compares the collision entropy with the Shannon entropy.

Remark 0.9 (Surprisal and the Fisher information). When the probabilities depend on a parameter θ, the surprisal of an outcome x is $-\log \mathbb{p}(x \mid \theta)$, which is minus the log-likelihood:

$$\mathbb{s}_\theta(x) = -\log \mathbb{p}(x \mid \theta) = -\ell(\theta).$$

Its derivative with respect to θ is minus the score:

$$\frac{\partial}{\partial\theta}\mathbb{s}_\theta(x) = -\ell'(\theta).$$

The expected squared score is the expected information, also called the Fisher information. It is also the expected second derivative of the surprisal, when the outcome X is drawn from the model at the true parameter value θ_0, and the score is evaluated at that same outcome X:

$$\mathcal{I}(\theta_0) = \mathbb{E}_{\theta_0}\left[\ell'(\theta_0)^2\right] = \mathbb{E}_{\theta_0}\left[\frac{\partial^2}{\partial\theta^2}\mathbb{s}_\theta(X)\Big|_{\theta=\theta_0}\right].$$

It is also the second derivative, at the true parameter value, of the Kullback–Leibler divergence between the model at that value and the model at the varying value (proved in the sds notes):

$$\frac{\partial^2}{\partial\theta^2} D_{\mathrm{KL}}(\theta_0 \,\|\, \theta)\Big|_{\theta=\theta_0} = \mathcal{I}(\theta_0).$$

Joint and conditional entropy

Definition 0.7 (Joint entropy)**.** Let X and Y be discrete random variables with joint probability mass function $p(x,y) \overset{\text{def}}{=} \mathbb{P}(X = x, Y = y)$. Their **joint entropy** is the entropy of the pair (X, Y):

$$H(X,Y) \overset{\text{def}}{=} -\sum_{x,y} p(x,y)\log p(x,y),$$

where the sum runs over the pairs with $p(x,y) > 0$.

Definition 0.8 (Entropy given one value)**.** For X and Y as in Definition 0.7, write $p(y \mid x) \stackrel{\text{def}}{=} \mathbb{P}(Y = y \mid X = x)$. The **entropy of Y given $X = x$**, for a value x with $\mathbb{P}(X = x) > 0$, is

$$H(Y \mid X = x) \stackrel{\text{def}}{=} -\sum_{y} p(y \mid x) \log p(y \mid x),$$

where the sum runs over the values y with $p(y \mid x) > 0$.

Definition 0.9 (Conditional entropy)**.** For X and Y as in Definition 0.7, write $p(x) \stackrel{\text{def}}{=} \mathbb{P}(X = x)$. The **conditional entropy** of Y given X averages the entropy of Y given each value of X over the distribution of X:

$$H(Y \mid X) \stackrel{\text{def}}{=} \sum_{x} p(x)\, H(Y \mid X = x),$$

where the sum runs over the values x with $p(x) > 0$.

Theorem 0.5 (Chain rule for entropy)**.** *For discrete random variables X and Y,*

$$H(X, Y) = H(X) + H(Y \mid X).$$

Proof

Proof. Use the notation of Definition 0.8 and Definition 0.9, with every sum over the pairs with $p(x, y) > 0$. Since $p(x, y) = p(x)\, p(y \mid x)$,

$$\begin{aligned}
H(X, Y) &= -\sum_{x,y} p(x, y) \log p(x, y) \\
&= -\sum_{x,y} p(x, y) \log[p(x)\, p(y \mid x)] \\
&= -\sum_{x,y} p(x, y) \log p(x) - \sum_{x,y} p(x, y) \log p(y \mid x) \\
&= -\sum_{x} \log p(x) \sum_{y} p(x, y) - \sum_{x} p(x) \sum_{y} p(y \mid x) \log p(y \mid x) \\
&= -\sum_{x} p(x) \log p(x) + \sum_{x} p(x)\, H(Y \mid X = x) \\
&= H(X) + H(Y \mid X).
\end{aligned}$$

The fourth line groups the terms by x and uses $p(x, y) = p(x)\, p(y \mid x)$ in the second sum; the fifth uses $\sum_y p(x, y) = p(x)$ and the definition of $H(Y \mid X = x)$. □

Corollary 0.2 (Chain rule for entropy, other order)**.** *For discrete random variables X and Y,*

$$\mathbb{H}(X, Y) = \mathbb{H}(X \mid Y) + \mathbb{H}(Y).$$

Proof

Proof. The pair (Y, X) takes each value (y, x) with the same probability that (X, Y) takes (x, y), so

$$\mathbb{H}(Y, X) = \mathbb{H}(X, Y).$$

The chain rule, applied with the roles of X and Y exchanged, gives

$$\mathbb{H}(Y, X) = \mathbb{H}(Y) + \mathbb{H}(X \mid Y).$$

Combining the two displays gives the claim. □

Corollary 0.3 (Conditional entropy as a difference)**.** *For discrete random variables X and Y with finite entropies,*

$$\mathbb{H}(X \mid Y) = \mathbb{H}(X, Y) - \mathbb{H}(Y).$$

Proof

Proof. Subtract $\mathbb{H}(Y)$ from both sides of Corollary 0.2:

$$\begin{aligned} \mathbb{H}(X, Y) - \mathbb{H}(Y) &= \mathbb{H}(X \mid Y) + \mathbb{H}(Y) - \mathbb{H}(Y) \\ &= \mathbb{H}(X \mid Y). \end{aligned}$$

□

Example 0.6 (Chain rule for entropy with a noisy copy)**.**
Let X be a fair coin toss, coded 0 or 1. Let Y be a noisy copy of X: it equals X with probability 0.8 and the other value with probability 0.2. The joint probabilities are 0.4 for $(0, 0)$ and $(1, 1)$, and 0.1 for $(0, 1)$ and $(1, 0)$. The chunk computes each entropy from its definition, in bits.

```r
entropy <- function(p) -sum(p * log2(p))
joint <- matrix(
  data = c(0.4, 0.1, 0.1, 0.4),
  nrow = 2,
  dimnames = list(x = 0:1, y = 0:1)
)
p_y <- colSums(joint)
cond_x_given_y <- sweep(x = joint, MARGIN = 2, STATS = p_y, FUN = "/")
h_x_given_y_values <- apply(X = cond_x_given_y, MARGIN = 2, FUN = entropy)
h_x_given_y <- sum(p_y * h_x_given_y_values)
h_xy <- entropy(joint)
h_y <- entropy(p_y)
c(h_xy = h_xy, h_x_given_y = h_x_given_y, h_y = h_y)
#>        h_xy h_x_given_y         h_y
#>   1.7219281   0.7219281   1.0000000
```

Here $\mathbb{H}(X, Y) = 1.722$ bits, $\mathbb{H}(X \mid Y) = 0.722$ bits and $\mathbb{H}(Y) = 1$ bits. So $\mathbb{H}(X \mid Y) + \mathbb{H}(Y) = 1.722$ bits agrees with $\mathbb{H}(X, Y)$ (Corollary 0.2), and $\mathbb{H}(X, Y) - \mathbb{H}(Y) = 0.722$ bits agrees with $\mathbb{H}(X \mid Y)$ (Corollary 0.3).

Definition 0.10 (Joint surprisal)**.** Let X and Y be discrete random variables with joint probability mass function $p(x, y) \stackrel{\text{def}}{=} \mathbb{P}(X = x, Y = y)$. The **joint surprisal** of a pair of outcomes (x, y) with $p(x, y) > 0$ is the surprisal of that pair:

$$\mathfrak{s}(x, y) \stackrel{\text{def}}{=} -\log p(x, y).$$

Definition 0.11 (Conditional surprisal)**.** For X and Y as in Definition 0.10, the **conditional surprisal** of an outcome y given an outcome x with $\mathbb{P}(X = x) > 0$ and $\mathbb{P}(Y = y \mid X = x) > 0$ is

$$\mathfrak{s}(y \mid x) \stackrel{\text{def}}{=} -\log \mathbb{P}(Y = y \mid X = x).$$

Example 0.7 (Joint and conditional surprisal of a noisy copy)**.**
Let X be a fair coin toss, coded 0 or 1. Let Y be a noisy copy of X: it equals X with probability 0.8 and the other value with probability 0.2. Take the pair of values $x = 0$ and $y = 1$. The joint probability is $\mathbb{P}(X = 0, Y = 1) = 0.1$, the probability of $X = 0$ is 0.5, and the conditional probability is $\mathbb{P}(Y = 1 \mid X = 0) = 0.2$. The chunk computes the three surprisals, in bits.

```r
s_xy <- -log2(0.1)
s_x <- -log2(0.5)
s_y_given_x <- -log2(0.2)
c(s_xy = s_xy, s_x = s_x, s_y_given_x = s_y_given_x)
#>        s_xy         s_x s_y_given_x
#>    3.321928    1.000000    2.321928
```

The joint surprisal is $\mathrm{s}(0,1) = 3.322$ bits, the surprisal of $x = 0$ alone is $\mathrm{s}(0) = 1$ bit, and the conditional surprisal is $\mathrm{s}(1 \mid 0) = 2.322$ bits.

Theorem 0.6 (Chain rule for surprisal)**.** *For discrete random variables X and Y and outcomes x and y with $\mathbb{P}(X = x, Y = y) > 0$,*

$$\mathrm{s}(x, y) = \mathrm{s}(x) + \mathrm{s}(y \mid x).$$

Proof

Proof. Since $\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\,\mathbb{P}(Y = y \mid X = x)$,

$$\begin{aligned} \mathrm{s}(x, y) &= -\log \mathbb{P}(X = x, Y = y) \\ &= -\log[\mathbb{P}(X = x)\,\mathbb{P}(Y = y \mid X = x)] \\ &= -\log \mathbb{P}(X = x) - \log \mathbb{P}(Y = y \mid X = x) \\ &= \mathrm{s}(x) + \mathrm{s}(y \mid x). \end{aligned}$$

□

Corollary 0.4 (Conditional surprisal as a difference)**.** *For X, Y, x and y as in Theorem 0.6,*

$$\mathrm{s}(y \mid x) = \mathrm{s}(x, y) - \mathrm{s}(x).$$

Proof

Proof. Subtract $\mathrm{s}(x)$ from both sides of Theorem 0.6:

$$\begin{aligned} \mathrm{s}(x, y) - \mathrm{s}(x) &= \mathrm{s}(x) + \mathrm{s}(y \mid x) - \mathrm{s}(x) \\ &= \mathrm{s}(y \mid x). \end{aligned}$$

□

Example 0.8 (Chain rule for surprisal with a noisy copy)**.**
For the pair of values in Example 0.7, the chunk recomputes the three surprisals in bits.

```
s_xy <- -log2(0.1)
s_x <- -log2(0.5)
s_y_given_x <- -log2(0.2)
c(sum = s_x + s_y_given_x, joint = s_xy)
#>      sum    joint
#> 3.321928 3.321928
```

The surprisal of $x = 0$ plus the conditional surprisal of $y = 1$ given $x = 0$ is 3.322 bits. This sum is the joint surprisal $\mathrm{s}(0, 1) = 3.322$ bits (Theorem 0.6). Subtracting $\mathrm{s}(0) = 1$ bit from the joint surprisal recovers $\mathrm{s}(1 \mid 0) = 2.322$ bits (Corollary 0.4).

Remark 0.10 (The chain rule for entropy is the chain rule for surprisal on average). The entropy of a pair is the expectation of its joint surprisal, and the conditional entropy is the expectation of the conditional surprisal:

$$\mathbb{H}(X, Y) = \mathbb{E}[\mathrm{s}(X, Y)]$$
$$\mathbb{H}(Y \mid X) = \mathbb{E}[\mathrm{s}(Y \mid X)].$$

Taking the expectation of both sides of Theorem 0.6 therefore gives Theorem 0.5.

Mutual information

Definition 0.12 (Mutual information)**.** The **mutual information** between discrete random variables X and Y with finite entropies is the reduction in the entropy of Y from learning X:

$$I(X; Y) \stackrel{\text{def}}{=} H(Y) - H(Y \mid X),$$

where $H(Y \mid X)$ is the conditional entropy.

Theorem 0.7 (Mutual information is symmetric)**.** *For discrete random variables X and Y with finite entropies,*

$$I(X; Y) = H(X) - H(X \mid Y) = I(Y; X).$$

Proof

Proof. The chain rule expands $H(X, Y)$ in either order:

$$H(X) + H(Y \mid X) = H(X, Y) = H(Y) + H(X \mid Y).$$

Subtracting $H(X \mid Y) + H(Y \mid X)$ from the outer two expressions gives

$$H(X) - H(X \mid Y) = H(Y) - H(Y \mid X).$$

The right side is $I(X; Y)$ by Definition 0.12, and the left side is $I(Y; X)$ by the same definition. □

Theorem 0.8 (Mutual information from the joint entropy)**.** *For discrete random variables X and Y with finite entropies,*

$$I(X; Y) = \mathbb{H}(X) + \mathbb{H}(Y) - \mathbb{H}(X, Y).$$

Proof

Proof. Subtract $\mathbb{H}(X)$ from both sides of the chain rule:

$$\mathbb{H}(Y \mid X) = \mathbb{H}(X, Y) - \mathbb{H}(X).$$

Substituting into Definition 0.12 gives

$$\begin{aligned} I(X; Y) &= \mathbb{H}(Y) - \mathbb{H}(Y \mid X) \\ &= \mathbb{H}(Y) - [\mathbb{H}(X, Y) - \mathbb{H}(X)] \\ &= \mathbb{H}(X) + \mathbb{H}(Y) - \mathbb{H}(X, Y). \end{aligned}$$

□

Theorem 0.9 (Mutual information from the conditional entropies)**.** *For discrete random variables X and Y with finite entropies,*

$$I(X; Y) = \mathbb{H}(X, Y) - [\mathbb{H}(X \mid Y) + \mathbb{H}(Y \mid X)].$$

Proof

Proof. By Corollary 0.3 and the chain rule, the entropy of each variable is the joint entropy minus a conditional entropy:

$$\mathbb{H}(X) = \mathbb{H}(X,Y) - \mathbb{H}(Y \mid X)$$
$$\mathbb{H}(Y) = \mathbb{H}(X,Y) - \mathbb{H}(X \mid Y).$$

Substituting both into Theorem 0.8 gives

$$\begin{aligned} I(X;Y) &= \mathbb{H}(X) + \mathbb{H}(Y) - \mathbb{H}(X,Y) \\ &= [\mathbb{H}(X,Y) - \mathbb{H}(Y \mid X)] + [\mathbb{H}(X,Y) - \mathbb{H}(X \mid Y)] - \mathbb{H}(X,Y) \\ &= \mathbb{H}(X,Y) - [\mathbb{H}(X \mid Y) + \mathbb{H}(Y \mid X)]. \end{aligned}$$

□

Theorem 0.10 (Decomposition of the sum of entropies)**.** *For discrete random variables X and Y with finite entropies,*

$$\mathbb{H}(X) + \mathbb{H}(Y) = \mathbb{H}(X \mid Y) + 2\, I(X;Y) + \mathbb{H}(Y \mid X).$$

Proof

Proof. Rearranging Theorem 0.9 gives

$$\mathbb{H}(X,Y) = I(X;Y) + \mathbb{H}(X \mid Y) + \mathbb{H}(Y \mid X).$$

Rearranging Theorem 0.8 gives

$$\mathbb{H}(X) + \mathbb{H}(Y) = I(X;Y) + \mathbb{H}(X,Y).$$

Substituting the first display into the second gives

$$\begin{aligned} \mathbb{H}(X) + \mathbb{H}(Y) &= I(X;Y) + [I(X;Y) + \mathbb{H}(X \mid Y) + \mathbb{H}(Y \mid X)] \\ &= \mathbb{H}(X \mid Y) + 2\, I(X;Y) + \mathbb{H}(Y \mid X). \end{aligned}$$

□

Theorem 0.11 (Twice the mutual information)**.** *For discrete random variables X and Y with finite entropies,*

$$2\, I(X;Y) = [\mathbb{H}(X) - \mathbb{H}(X \mid Y)] + [\mathbb{H}(Y) - \mathbb{H}(Y \mid X)].$$

Proof

Proof. By Theorem 0.7,

$$\mathbb{H}(X) - \mathbb{H}(X \mid Y) = I(X;Y)$$
$$\mathbb{H}(Y) - \mathbb{H}(Y \mid X) = I(X;Y).$$

Adding the two displays gives

$$[\mathbb{H}(X) - \mathbb{H}(X \mid Y)] + [\mathbb{H}(Y) - \mathbb{H}(Y \mid X)] = I(X;Y) + I(X;Y)$$
$$= 2\, I(X;Y).$$

□

Example 0.9 (Mutual information identities with a noisy copy)**.**
Take the fair coin X and its noisy copy Y from Example 0.6. The chunk computes the entropies from the joint probabilities in bits, then the mutual information in four ways.

```
entropy <- function(p) -sum(p * log2(p))
joint <- matrix(
  data = c(0.4, 0.1, 0.1, 0.4),
  nrow = 2
)
h_xy <- entropy(joint)
h_x <- entropy(rowSums(joint))
h_y <- entropy(colSums(joint))
h_x_given_y <- h_xy - h_y
h_y_given_x <- h_xy - h_x
mi_definition <- h_y - h_y_given_x
mi_joint <- h_x + h_y - h_xy
mi_conditionals <- h_xy - (h_x_given_y + h_y_given_x)
mi_twice <- ((h_x - h_x_given_y) + (h_y - h_y_given_x)) / 2
c(mi_definition, mi_joint, mi_conditionals, mi_twice)
#> [1] 0.2780719 0.2780719 0.2780719 0.2780719
```

All four agree: $I(X;Y) = 0.278$ bits (Theorem 0.8, Theorem 0.9 and Theorem 0.11). Also $\mathbb{H}(X) + \mathbb{H}(Y) = 2$ bits matches $\mathbb{H}(X \mid Y) + 2\, I(X;Y) + \mathbb{H}(Y \mid X) = 2$ bits (Theorem 0.10).

Remark 0.11 (Entropies as areas). The identities above match a picture of two overlapping regions, with the area of one region standing for $\mathbb{H}(X)$ and the area of the other for $\mathbb{H}(Y)$. The overlap has area $I(X;Y)$ (Theorem 0.8). The part of the first region outside the overlap has area $\mathbb{H}(X \mid Y)$, and the part of the second region outside the overlap has area $\mathbb{H}(Y \mid X)$. The two regions together have area $\mathbb{H}(X, Y)$ (Theorem 0.9). The picture is a mnemonic for two variables only.

Remark 0.12 (Mutual information as a divergence). Cover and Thomas (2006, sec. 2.3) define mutual information as the Kullback–Leibler divergence of the joint distribution of (X, Y) from the product of its marginals, and show that this equals $H(Y) - H(Y \mid X)$ (Cover and Thomas 2006, sec. 2.4). Since a Kullback–Leibler divergence is never negative, and is 0 only when the two distributions are equal (Cover and Thomas 2006, sec. 2.6), $I(X;Y) \geq 0$, with equality exactly when X and Y are independent. Learning X never raises the entropy of Y on average.

Definition 0.13 (Information gain)**.** A decision tree splits the points at a node into branches. The **information gain** of the split is the entropy of the class labels at the node, minus the entropy of the labels in each branch, weighted by the share of the node's points that branch holds (Quinlan 1986).

Remark 0.13 (Information gain is mutual information). Draw one point uniformly at random from the node of Definition 0.13, and let Y be its class and X the branch it falls into. Then the entropy at the node is $H(Y)$, the weighted entropy left in the branches is $H(Y \mid X)$, and the information gain is the mutual information $I(X;Y)$.

Exercise 0.1 (Choosing between two splits)**.** A node holds 20 points, 10 red and 10 green. Two candidate splits each send the points into two branches:

split	branch	red	green
A	yes	4	0
	no	6	10
B	yes	5	10
	no	5	0

Draw one point uniformly at random from the node, let Y be its color, and let X be the branch it falls into (Remark 0.13). The information gain of each split (Definition 0.13) is then $I(X;Y)$. Use base-2 logarithms throughout.

1. Compute $H(Y)$.
2. For each split, compute $H(Y \mid X = x)$ for both branches, and then $H(Y \mid X)$.
3. Compute the information gain $I(X;Y)$ for each split. Which split has the larger gain?

Solution

Solution 0.1.

1. Half the points are red, so

$$\begin{aligned} H(Y) &= -\tfrac{1}{2}\log_2 \tfrac{1}{2} - \tfrac{1}{2}\log_2 \tfrac{1}{2} \\ &= 1 \text{ bit.} \end{aligned}$$

2. A branch holding one color only has entropy 0. For split A, the "no" branch holds 6 red and 10 green points of 16:

$$\begin{aligned} H(Y \mid X = \text{no}) &= -\tfrac{6}{16}\log_2 \tfrac{6}{16} - \tfrac{10}{16}\log_2 \tfrac{10}{16} \\ &\approx 0.531 + 0.424 \\ &\approx 0.954 \text{ bits.} \end{aligned}$$

   The "yes" branch holds 4 of the 20 points and the "no" branch 16, so

$$\begin{aligned} H(Y \mid X) &= \tfrac{4}{20} \cdot 0 + \tfrac{16}{20} \cdot 0.954 \\ &\approx 0.764 \text{ bits.} \end{aligned}$$

   For split B, the "yes" branch holds 5 red and 10 green points of 15:

$$\begin{aligned} H(Y \mid X = \text{yes}) &= -\tfrac{1}{3}\log_2 \tfrac{1}{3} - \tfrac{2}{3}\log_2 \tfrac{2}{3} \\ &\approx 0.528 + 0.390 \\ &\approx 0.918 \text{ bits.} \end{aligned}$$

   The "yes" branch holds 15 of the 20 points and the "no" branch 5, so

$$\begin{aligned} H(Y \mid X) &= \tfrac{15}{20} \cdot 0.918 + \tfrac{5}{20} \cdot 0 \\ &\approx 0.689 \text{ bits.} \end{aligned}$$

3. By Definition 0.12, the gain is $H(Y) - H(Y \mid X)$:

   - split A: $1 - 0.764 = 0.236$ bits
   - split B: $1 - 0.689 = 0.311$ bits

   Split B has the larger gain. Both splits isolate a branch of one color, so the gain depends on the other, mixed branch. B's mixed branch, 5 red against 10 green, is more lopsided than A's 6 against 10, so it has less entropy, and it also holds fewer of the points, 15 against 16, so its entropy gets less weight.

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